

NumPy Functions Cheat Sheet

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1 Array Creation

```
import numpy as np

np.array([1,2,3])
np.zeros((3,4))
np.ones((2,5))
np.empty((2,2))
np.full((3,3),10)
np.eye(4)
np.identity(3)
np.arange(1,11)
np.arange(0,20,2)
np.linspace(0,10,5)
np.logspace(1,3,5)
```

2 Random Functions

```
np.random.rand(3,3)
np.random.randn(3,3)
np.random.randint(1,100,(3,4))
np.random.choice([10,20,30],5)
np.random.seed(42)
np.random.shuffle(arr)
np.random.permutation(arr)
```

3 Array Information

```
arr.ndim
arr.shape
arr.size
arr.dtype
arr.itemsize
arr.nbytes
```

4 Reshaping Arrays

```
arr.reshape(3,4)
arr.flatten()
arr.ravel()
arr.T
arr.transpose()
```

5 Indexing and Slicing

```
arr[2]  
arr[1:5]  
arr[:,1]  
arr[1,:]  
arr[1:3,2:4]  
arr[::-1]
```

6 Mathematical Functions

```
np.add(a,b)  
np.subtract(a,b)  
np.multiply(a,b)  
np.divide(a,b)  
np.power(a,2)  
np.sqrt(a)  
np.square(a)  
np.abs(a)  
np.mod(a,2)
```

7 Statistical Functions

```
np.sum(arr)  
np.mean(arr)  
np.median(arr)  
np.std(arr)  
np.var(arr)  
np.min(arr)  
np.max(arr)  
np.argmin(arr)  
np.argmax(arr)  
np.percentile(arr,50)
```

8 Cumulative Functions

```
np.cumsum(arr)  
np.cumprod(arr)
```

9 Trigonometric Functions

```
np.sin(arr)
np.cos(arr)
np.tan(arr)
np.arcsin(arr)
np.arccos(arr)
np.arctan(arr)
np.degrees(arr)
np.radians(arr)
```

10 Exponential and Logarithmic Functions

```
np.exp(arr)
np.log(arr)
np.log10(arr)
np.log2(arr)
```

11 Sorting and Searching

```
np.sort(arr)
np.argsort(arr)
np.where(arr>10)
np.searchsorted(arr,5)
```

12 Unique Values

```
np.unique(arr)
np.unique(arr,return_counts=True)
```

13 Joining Arrays

```
np.concatenate((a,b))
np.vstack((a,b))
np.hstack((a,b))
np.column_stack((a,b))
np.row_stack((a,b))
```

14 Splitting Arrays

```
np.split(arr,2)
np.array_split(arr,3)
np.vsplit(arr,2)
np.hsplit(arr,2)
```

15 Insert, Append and Delete

```
np.insert(arr,2,100)
np.append(arr,50)
np.delete(arr,3)
```

16 Boolean Masking

```
arr[arr>10]
arr[(arr>5)&(arr<20)]
np.logical_and(a,b)
np.logical_or(a,b)
```

17 Linear Algebra

```
np.dot(a,b)
np.matmul(a,b)
A @ B
np.linalg.det(A)
np.linalg.inv(A)
np.linalg.eig(A)
np.linalg.solve(A,b)
```

18 Broadcasting

```
arr+10
arr*5
arr+np.array([1,2,3])
```

19 Copy Functions

```
arr.copy()
arr.view()
```

20 Handling Missing Values

```
np.isnan(arr)
np.isinf(arr)
np.isfinite(arr)
np.nanmean(arr)
np.nansum(arr)
```

21 Conditional Functions

```
np.where(arr>10,1,0)
np.clip(arr,0,100)
```

22 Iteration

```
for x in np.nditer(arr):
    print(x)

for index,value in np.ndenumerate(arr):
    print(index,value)
```

23 File Operations

```
np.save("data.npy",arr)
np.load("data.npy")
np.savetxt("data.csv",arr,delimiter=",")
np.loadtxt("data.csv",delimiter=",")
```

24 Utility Functions

```
np.all(arr)
np.any(arr)
np.allclose(a,b)
np.array_equal(a,b)
np.repeat(arr,2)
np.tile(arr,3)
np.flip(arr)
np.roll(arr,2)
np.diag(arr)
np.trace(arr)
np.meshgrid(x,y)
```

25 Top 30 NumPy Functions

Function	Purpose
array()	Create NumPy array
zeros()	Array filled with zeros
ones()	Array filled with ones
full()	Fill array with constant value
eye()	Identity matrix
arange()	Range of values
linspace()	Evenly spaced values
reshape()	Change shape of array
flatten()	Flatten array
ravel()	Flatten without copy if possible
sum()	Sum of elements
mean()	Average value
median()	Median value
std()	Standard deviation
var()	Variance
min()	Minimum value
max()	Maximum value
argmin()	Index of minimum
argmax()	Index of maximum
sort()	Sort array
unique()	Unique elements
concatenate()	Join arrays
vstack()	Vertical stack
hstack()	Horizontal stack
split()	Split arrays
where()	Conditional selection
dot()	Dot product
matmul()	Matrix multiplication
transpose()	Transpose matrix
random.randint()	Generate random integers